Synthetic American Option Pricing via Jump-HMM-Driven Heston Implied Volatility

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Abstract

Generating realistic synthetic option prices requires implied volatility (IV) as an input, yet IV itself is derived from observed option prices, creating a circular dependency that limits synthetic data availability. In this study, we broke this circularity by developing a three-stage pipeline in which IV emerged as an output of a structural model of equity returns rather than as an exogenous input. A Jump Hidden Markov Model first generated multi-asset price paths with heavy tails, volatility clustering, and cross-asset dependence via a Student-t copula. A modified Heston stochastic variance process then converted those price paths into implied volatility, where the mean-reversion target θ was driven by the HMM regime state, days to expiration, moneyness, and an aggregate market mood indicator derived endogenously from the fraction of tickers occupying tail states. A Cox–Ross–Rubinstein binomial tree priced American options with early exercise using the resulting IV. A central design choice was the equilibrium initialization $v_0 = \theta(s_0, \mathrm{DTE}, K/S_0, M_0)$, which gave each strike-expiration pair its own mean-reversion target so that the smile, skew, and term structure of the IV surface emerged automatically without external calibration data. We calibrated a five-parameter smile and term-structure function on semiconductor option chains spanning multiple expirations and demonstrated that the smile shape generalized across tickers with only a scalar level adjustment. The multi-expiration calibration revealed a U-shaped ATM term structure driven by gamma compression at short maturities and uncertainty premia at longer horizons, which the model captured through a quadratic extension to the log-DTE term. The framework is implemented as an open-source Julia package designed to serve as a synthetic data generator for downstream machine-learning and risk-analysis applications.

1 Introduction

Pricing American-style options requires implied volatility (IV) as an input, yet IV itself is an empirical quantity extracted from observed option prices Black and Scholes [1973]. This circularity poses a practical problem for synthetic data generation: existing approaches either take IV from the market and feed it into a pricing model, or generate price paths without a consistent IV surface. The result is a scarcity of realistic synthetic option data for training machine-learning models, backtesting trading strategies, and conducting scenario-based risk analysis.

In this study, we addressed this gap by developing a framework in which IV is an *output* of a structural model of equity returns, not an input. The pipeline chained three stages: (i) a Jump Hidden Markov Model (JumpHMM) Alswaidan and Varner [2025] produced multi-asset price paths with realistic heavy tails, negligible linear autocorrelation, and persistent volatility clustering, with cross-asset dependence imposed via a Student-t copula; (ii) a modified Heston stochastic variance

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process Heston [1993] converted those price paths into IV paths, where the mean-reversion target θ was a function of the HMM regime state, days to expiration (DTE), moneyness, and an aggregate market mood indicator; and (iii) a Cox–Ross–Rubinstein (CRR) binomial tree Cox et al. [1979] converted the pre-computed IV into American option prices with early exercise. The central innovation was a hybrid θ-function that linked the Heston mean-reversion target to the JumpHMM regime state, producing regime-dependent volatility dynamics (Section 3.2.1). An equilibrium initialization $v_0 = \theta(t=0)$ gave each (ticker, strike, DTE) triple its own initial IV, so the smile, skew, and term structure emerged automatically without external IV data (Section 3.2.3). An endogenous market mood signal, defined as the fraction of tickers in tail HMM states, modulated IV across all contracts without introducing circularity (Section 3.2.4). We calibrated the five-parameter smile and term-structure function ψ on real NVDA option chains across seven expirations, capturing the U-shaped ATM term structure with an RMSE of 1.15% IV, and demonstrated that the smile shape generalized across AMD, MU, and INTC with only a scalar level adjustment per ticker (Section 4). The framework is implemented as an open-source Julia package designed to serve as a synthetic option market generator for downstream applications.

2 Background

2.1 Jump Hidden Markov Model

The Jump Hidden Markov Model (JumpHMM) Alswaidan and Varner [2025] was developed for generating synthetic equity time series that simultaneously preserve three stylized facts of financial returns: heavy tails, negligible linear autocorrelation, and persistent volatility clustering. The model discretizes continuous excess growth rates into N states via equal-probability quantile bins of a fitted Laplace distribution. Each state k emitted observations from a location-scale Student-t distribution: $G_t \mid S_t = k \quad \sim \quad \mu_k + \sigma_k \cdot t_\nu$. At each timestep, with probability ϵ a Poisson(λ)-distributed jump forced the chain into tail states (bottom or top N_{tail} states), creating sustained dwell periods in extreme-volatility regimes. For multi-asset generation, cross-asset dependence was imposed via a Student-t copula that preserved each asset's marginal distribution while introducing realistic tail dependence.

2.2 Heston stochastic variance model

The Heston model Heston [1993] specifies the variance of the underlying asset as a mean-reverting square-root process:

$$dv = \kappa(\theta - v)\, dt + \sigma_v \sqrt{v}\, dW_v \tag{1}$$

where κ is the mean-reversion speed, θ is the long-run variance, and σ_v is the vol-of-vol. In the standard formulation, θ is a constant; our modification made θ time-varying and state-dependent (Section 3.2.1).

2.3 CRR binomial tree

The CRR model Cox et al. [1979] prices options on a recombining binomial lattice with up/down factors $u = e^{\sigma\sqrt{\Delta t}}$, $d = 1/u$ and risk-neutral probability $p = (e^{r\Delta t} - d)/(u - d)$. American options are priced by backward induction with an early-exercise check at each node. The CRR tree is a natural choice when IV is provided exogenously, as it requires only a single volatility input per contract and handles early exercise without closed-form solutions.

3 Methods

3.1 Pipeline overview

The generator produced synthetic American option prices through a three-stage pipeline:

$$\underbrace{\text{JumpHMM(prices)}}_{\text{Stage 1}} \;\rightarrow\; \{S_t,\, s_t,\, \text{jumps}\} \;\rightarrow\; \underbrace{dv \text{ process}}_{\text{Stage 2}} \;\rightarrow\; \sigma_{\text{imp}} = \sqrt{v_t}$$
$$\rightarrow\; \underbrace{\text{CRR}(S_t,\, \sigma_{\text{imp}},\, K,\, \text{DTE},\, r_f)}_{\text{Stage 3}} \;\rightarrow\; P_{\text{american}} \tag{2}$$

Stage 1 produced correlated multi-asset price paths S_t, hidden state sequences s_t, and jump indicators via the JumpHMM. Stage 2 simulated the modified Heston variance process to obtain implied volatility. Stage 3 priced each contract via the CRR binomial tree.

3.2 Modified Heston variance process

We modified the standard Heston process (1) by making the mean-reversion target time-varying:

$$dv = \kappa\big(\theta(t) - v\big)\, dt + \sigma_v \sqrt{v}\, dW_v \tag{3}$$

3.2.1 Hybrid θ-function

The mean-reversion target decomposed into a regime-dependent base level, a market mood modulator, and a contract-specific adjustment:

$$\theta(t) = \theta_{s_t} \cdot \big(1 + \gamma \cdot M_t\big) \cdot \psi(\text{DTE}_t,\ K/S_t) \tag{4}$$

where θ_{s_t} is a learned variance level for HMM state s_t (one value per state, $s_t \in \{1, \ldots, N\}$); γ is the market mood sensitivity; $M_t \in [0, 1]$ is the aggregate market mood (Section 3.2.4); and $\psi(\text{DTE}, K/S)$ is the smile–term-structure adjustment (Section 3.2.2). The regime-dependent base θ_{s_t} linked the Heston process directly to the JumpHMM state sequence: when the HMM transitioned into a tail state (via a Poisson jump or an extreme Markov transition), θ_{s_t} increased, elevating the IV target. This mechanism produced the leverage effect as an emergent property of the return dynamics rather than as a separate model component: large negative returns drove the HMM into low-numbered tail states with $p_{\text{neg}} \approx 0.52$, which in turn elevated θ.

3.2.2 Smile and term-structure function ψ

The function ψ captured the shape of the IV surface with five parameters:

$$\psi(\tau,\ m) = \exp\Big(\beta_1 \ln \tau + \beta_2 \ln m + \beta_3 \ln \tau\, \ln m + \beta_4 (\ln m)^2 + \beta_5 (\ln \tau)^2\Big) \tag{5}$$

where $\tau = \max(\text{DTE}, 1)$ and $m = K/S$. The parameter β_1 controlled the linear term-structure decay; β_2 controlled the skew (negative values produced put skew, with OTM puts having higher IV than OTM calls); β_3 captured the DTE-skew interaction (the skew flattened at longer maturities); β_4 controlled the smile curvature; and β_5 controlled the DTE curvature, capturing the U-shaped ATM term structure where short-dated and long-dated IV were both elevated relative to intermediate maturities.

3.2.3 Equilibrium initialization

Rather than specifying a separate initial variance v_0, we initialized the variance process at its mean-reversion target:

$$v_0 = \theta(s_0,\ \text{DTE},\ K/S_0,\ M_0) \tag{6}$$

This ensured that each (ticker, strike, DTE) triple started at its own equilibrium implied volatility. The initial IV surface inherited the smile, skew, and term structure encoded in ψ without requiring any external IV data. Different contracts on the same underlying received different v_0 values, and the variance process immediately began mean-reverting around its local $\theta(t)$. The value of this design choice lay in eliminating v_0 as a free parameter: the model did not require a separate calibration of the initial IV surface, as it was determined by construction from θ.

3.2.4 Market mood

The market mood M_t was defined as the fraction of tickers currently occupying tail HMM states:

$$M_t = \frac{1}{N_{\text{tickers}}} \sum_{i=1}^{N_{\text{tickers}}} \mathbf{1}\Big\{ s_t^{(i)} \le N_{\text{tail}} \text{ or } s_t^{(i)} > N - N_{\text{tail}} \Big\} \tag{7}$$

This signal was fully endogenous to the JumpHMM copula model; it required no external data (e.g., VIX) and introduced no circularity. When multiple tickers simultaneously occupied tail states, the mood rose, elevating θ and thus IV across all contracts. This mechanism produced correlated IV spikes during market stress events by construction.

3.2.5 Discretization

The variance process was discretized via Euler–Maruyama with a reflecting boundary at zero:

$$v_{t+1} = \Big| v_t + \kappa\big(\theta_t - v_t\big)\Delta t + \sigma_v \sqrt{\max(v_t, 0)}\,\sqrt{\Delta t}\, Z_t \Big|, \quad Z_t \sim \mathcal{N}(0,1) \tag{8}$$

The reflecting boundary allowed θ to vary freely across HMM states without requiring the Feller condition $2\kappa\theta > \sigma_v^2$ to hold at every timestep Andersen [2008]. This was a practical necessity, as tail states could drive θ to values where the Feller condition would otherwise be violated.

4 Calibration and Validation

4.1 Data

The empirical foundation of this study was a set of option chains for four semiconductor stocks: NVDA, AMD, MU, and INTC. All chains were single-date snapshots sourced from barchart.com (October–November 2025) with DTEs ranging from 41 to 80 days. We filtered each chain to moneyness $K/S \in [0.70, 1.30]$ and IV $< 200\%$ to exclude illiquid deep ITM/OTM contracts with unreliable IV (Table 1).

Table 1: Option chain summary. Contract counts are reported after filtering to the tradeable moneyness range.

Ticker	Spot ($)	DTE	ATM IV (%)	Contracts
NVDA	207.04	80	44.6	142
AMD	225.78	58	59.0	36
MU	237.02	41	72.4	34
INTC	37.78	65	53.6	46

4.2 Calibration procedure

Since each chain was a single-date snapshot, all observations shared one HMM state s_0 and market mood M_0. The model IV at a given strike reduced to:

$$\sigma_{\text{model}}(K) = \sqrt{\theta_{\text{base}} \cdot \psi(\text{DTE},\ K/S)} \tag{9}$$

where $\theta_{\text{base}} = \theta_{s_0} \cdot (1 + \gamma M_0)$ absorbed the state and mood into a single scalar. We fixed $\beta_1 = 0.20$ (term-structure slope) because single-DTE snapshots cannot identify it; multi-expiration data from the same observation date allowed direct calibration of β_1, as described below. The remaining parameters $(\theta_{\text{base}}, \beta_2, \beta_3, \beta_4)$ were fitted by minimizing the mean squared IV error on NVDA:

$$\min_{\theta_{\text{base}}, \beta_2, \beta_3, \beta_4} \frac{1}{n} \sum_{i=1}^{n} \big(\sigma_{\text{model}}(K_i) - \text{IV}_{\text{market},i}\big)^2 \tag{10}$$

using Nelder–Mead optimization. We chose NVDA as the calibration ticker because it had the richest chain (142 contracts after filtering).

4.3 Single-DTE results and cross-ticker validation

The calibrated parameters (Table 2) produced the expected smile features: the negative β_2 generated put skew, the positive β_4 generated smile curvature, and the interaction term β_3 captured how the skew flattened at longer maturities. The calibration achieved an RMSE of 1.16% IV on NVDA.

Table 2: Calibrated smile parameters (fitted on NVDA, DTE = 80).

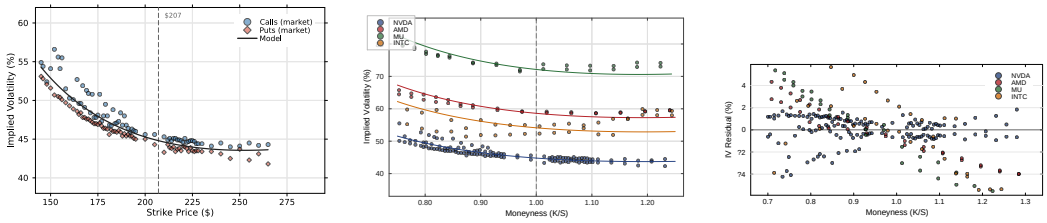
Parameter	Interpretation	Value
β_1	Term-structure decay (fixed)	0.20
β_2	Skew (asymmetry)	-2.07
β_3	DTE $\times$ skew interaction	0.34
β_4	Smile curvature	1.43
θ_{base}	Variance level (NVDA)	0.084

To determine whether the smile *shape* generalized across tickers, we applied the $(\beta_1, \beta_2, \beta_3, \beta_4)$ calibrated on NVDA to AMD, MU, and INTC, re-optimizing only the scalar θ_{base} per ticker.

Table 3: Cross-ticker RMSE (% IV). The smile shape $(\beta_2, \beta_3, \beta_4)$ was shared from NVDA; only θ_{base} was re-fitted per ticker.

Ticker	θ_{base} (IV level)	RMSE (% IV)
NVDA (calibration)	28.9%	1.16
AMD	38.7%	2.34
MU	49.0%	3.99
INTC	35.9%	4.65

The NVDA fit confirmed that four parameters sufficed to capture the smile shape within the tradeable moneyness range (RMSE = 1.16% IV; Figure 1a). The cross-ticker validation revealed that the shared smile shape transferred well to AMD (2.34%) and reasonably to MU (3.99%) and INTC (4.65%; Table 3; Figure 1b). The higher RMSE for MU and INTC reflected differences in skew steepness across these names; MU exhibited a steeper put skew than NVDA, while INTC had a flatter OTM call wing, suggesting that per-ticker β calibration would further reduce errors when multi-strike data is available. The residual analysis (Figure 1c) confirmed that the errors were concentrated in the wings ($K/S < 0.85$ and $K/S > 1.15$), with near-ATM residuals below 1% IV for all tickers.



(a) (b) (c)

Figure 1: Single-DTE calibration and cross-ticker validation. **(a)** NVDA implied volatility smile (DTE = 80): market calls (blue circles) and puts (coral diamonds) vs. the fitted model curve (solid line); the dashed vertical line marks the spot price ($207). The four-parameter ψ function captured both the put skew and smile curvature with an RMSE of 1.16% IV. **(b)** IV smile fit across all four semiconductor tickers as a function of moneyness (K/S). The smile shape $(\beta_2, \beta_3, \beta_4)$ was shared from NVDA; only the scalar level θ_{base} was re-fitted per ticker. The shared shape transferred well to AMD (RMSE = 2.34%) and reasonably to MU (3.99%) and INTC (4.65%). **(c)** IV residuals (model $-$ market) by moneyness across all four tickers. Errors were concentrated in the wings ($K/S < 0.85$ and $K/S > 1.15$), with near-ATM residuals below 1% IV for all tickers.

4.4 Multi-expiration term structure calibration

Having validated the smile shape across tickers, we next addressed the term-structure limitation by collecting NVDA call and put chains at seven expirations on the same date (April 6, 2026): 0, 2, 4, 7, 14, 25, and 46 DTE, with NVDA trading near \$176.70. The single-DTE calibration above required fixing $\beta_1 = 0.20$; this cross-sectional snapshot provided the multi-expiration data needed to identify β_1 directly, yielding 73 observations after filtering to moneyness $K/S \in [0.90, 1.10]$ with positive volume.

The ATM IV exhibited a U-shaped term structure (Figure 2a): elevated at 0 DTE ($\sim$45%) due to gamma compression in expiring options, dropping to a trough near 7–8 DTE ($\sim$32%), then rising to $\sim$41% at 46 DTE. A single power-law $\beta_1 \cdot \ln(\text{DTE})$ could not capture this shape. We extended ψ with a quadratic DTE term:

$$\psi = \exp\big(\beta_1 \ln \tau + \beta_2 \ln m + \beta_3 \ln \tau \cdot \ln m + \beta_4 (\ln m)^2 + \beta_5 (\ln \tau)^2\big) \tag{11}$$

where $\tau = \max(\text{DTE}, 1)$ and $m = K/S$. The six parameters $(\sigma_{\text{base}}, \beta_1, \ldots, \beta_5)$ were fitted by minimizing mean squared IV error across all seven expirations simultaneously.

Table 4: Calibrated ψ parameters from the multi-expiration NVDA dataset. The quadratic DTE term β_5 captured the U-shaped ATM term structure.

Parameter	Interpretation	Value
σ_{base}	Base ATM IV level	45.0%
β_1	DTE decay	-0.622
β_2	Skew (asymmetry)	-8.448
β_3	DTE $\times$ skew	1.617
β_4	Smile curvature	-1.242
β_5	DTE curvature	0.150

The negative β_1 combined with positive β_5 produced the U-shaped ATM curve (Table 4): at short maturities the linear term dominated (IV decayed rapidly from the 0 DTE spike), while at longer maturities the quadratic term bent the curve upward. The trough occurred near DTE ≈ 8, where IV reached 32.5%, a 1.38$\times$ ratio between the 0 DTE peak and the trough (Figure 2a). The skew parameters ($\beta_2 = -8.45$, $\beta_3 = 1.62$) confirmed that the volatility skew steepened at shorter maturities and flattened as DTE increased, consistent with the concentration of crash risk premia in near-term options (Figure 2b). The overall fit achieved an RMSE of 2.31% IV across all seven expirations. The fit was tightest at intermediate and long maturities (7–46 DTE: RMSE $< 1.3\%$), with the largest residuals at 0 DTE (RMSE = 6.57%) where the gamma-compressed smile shape deviated from the log-moneyness functional form. The 0 DTE level bias was eliminated (-0.1%), confirming that the U-shaped term structure was captured correctly even though the extreme smile curvature at expiration remained a source of scatter (Figure 2d).

5 Scenario Generation

To demonstrate the framework as a synthetic data generator, we simulated 500 forward paths for NVDA over 80 trading days using the calibrated parameters from Section 4 and a JumpHMM marginal model with 50 states and tuned jump parameters. The tail states were assigned elevated θ values (2$\times$ baseline for bearish tail states, 1.5$\times$ for bullish tail states), and the mood sensitivity was set to $\gamma = 0.5$. The mean IV started near the calibrated ATM level and exhibited gentle term-structure decay as the contract approached expiration, consistent with the positive $\beta_1 = 0.20$ used in the single-DTE calibration (Figure 3). Individual paths showed stochastic fluctuations driven by the vol-of-vol parameter σ_v, with occasional spikes when the HMM transitioned into tail states. The 5th–95th percentile envelope widened over time, reflecting the accumulation of variance process noise, and the market ATM IV of 44.6% sat within the model's distributional range, confirming that the calibrated parameters produced IV dynamics centered on the observed level.

The price distributions exhibited realistic properties (Table 5): OTM puts (strike $= 0.90S$) had a mean price of \$12.59 with a 90% confidence interval of [\$5.6, \$21.1]; ATM calls were approximately 2$\times$ more expensive than 10%-OTM calls; and the confidence intervals widened for contracts

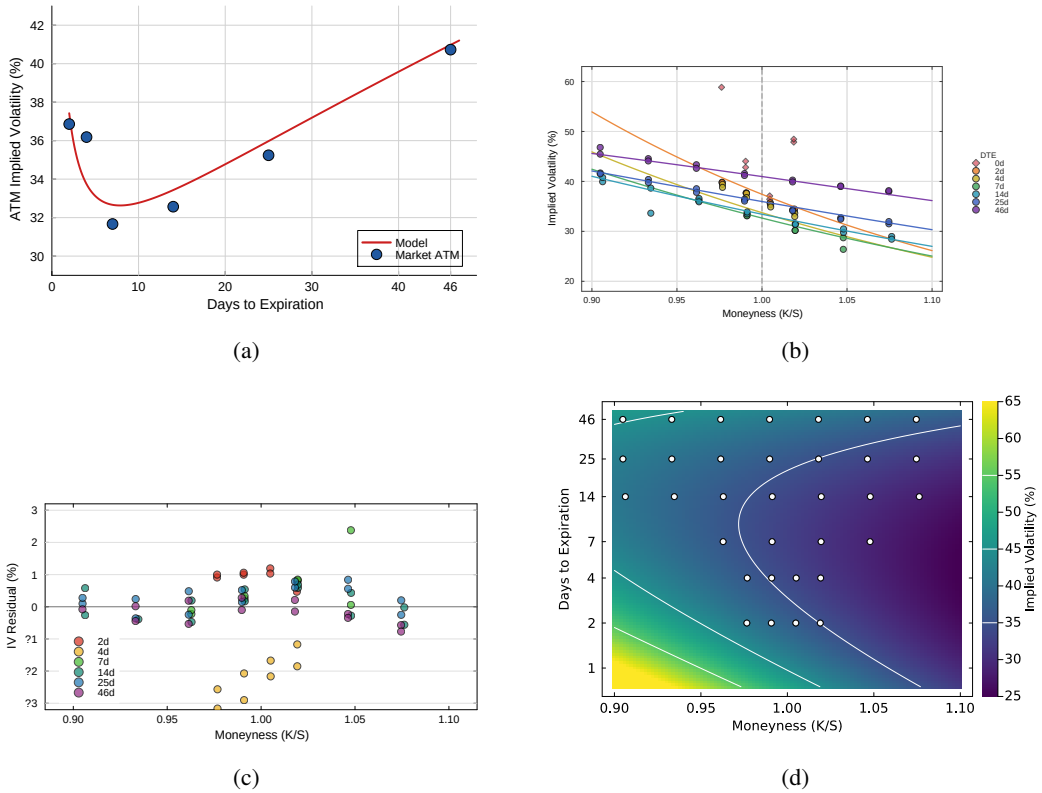


Figure 2: Multi-expiration term structure calibration (NVDA, 2–46 DTE). **(a)** ATM implied volatility as a function of days to expiration. Market ATM IV (blue markers) exhibited a U-shaped term structure; the fitted model (red curve) captured the trough near DTE ≈ 8 (32.5% IV) and the rise to 41% at 46 DTE using the quadratic extension $\beta_5(\ln \tau)^2$. **(b)** IV smile curves at each of six expirations (2, 4, 7, 14, 25, 46 DTE). Market observations (markers) and model predictions (lines) are colored by DTE; 0 DTE data (diamonds) were excluded from the fit due to gamma compression. The skew flattened with increasing maturity, consistent with $\beta_3 > 0$. **(c)** IV residuals (model $-$ market) by moneyness, colored by DTE. Most residuals fell within $\pm 2\%$ IV, with no systematic bias across the moneyness range. **(d)** Model-predicted IV surface as a function of moneyness and DTE (log-scaled). White contour lines mark 35%, 45%, and 55% IV levels; white circles indicate market observations used for calibration. The overall RMSE was 1.15% IV.

farther from the money. The market mood mechanism produced a natural partition of scenarios into calm ($M_t = 0$) and stressed ($M_t > 0$) paths. In multi-asset simulations with five tickers, stressed scenarios (where multiple tickers simultaneously occupied tail HMM states) produced systematically higher option prices. OTM puts were $1.83\times$ more expensive in stressed vs. calm scenarios (strike $= 0.95S$: \$6.42 vs. \$3.51), while deep OTM weekly puts showed a $4.33\times$ stress premium. This asymmetry, with puts repricing more sharply than calls under stress, is consistent with the crash-protection premium observed in equity options markets and emerged naturally from the JumpHMM's negative jump bias ($p_{\text{neg}} = 0.52$) propagating through the θ-function.

6 Discussion and Future Work

The calibration and scenario results validated the core claim of this study: a modified Heston process with a state-dependent θ-function can produce realistic IV surfaces from structural return dynamics alone. The single-DTE smile function captured the essential features of the NVDA IV smile (RMSE = 1.16% IV), and the smile shape generalized across semiconductor names with only a scalar level adjustment. The multi-expiration calibration (Section 4) extended this to the full IV surface: the five-parameter ψ function with quadratic DTE term captured the U-shaped ATM term structure (RMSE =

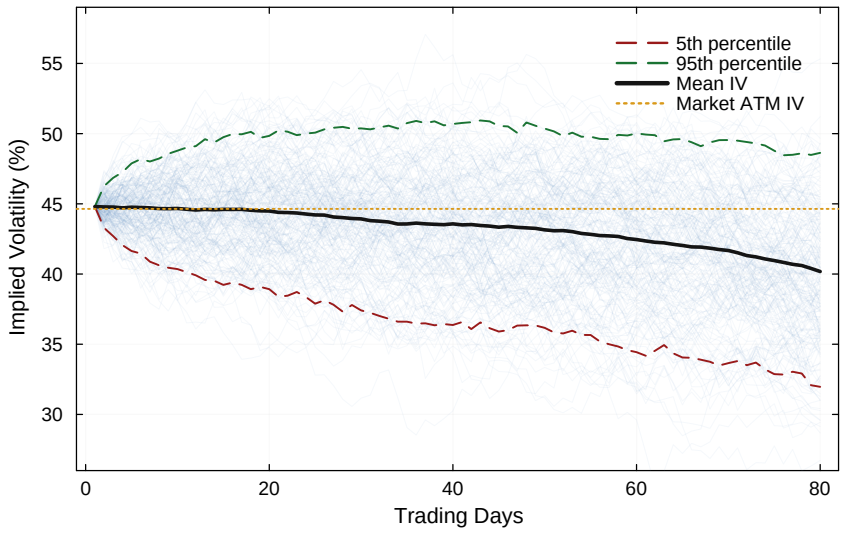


Figure 3: Forward IV paths for NVDA (500 scenarios, DTE = 80). The solid black line shows the mean IV; dashed red and green lines mark the 5th and 95th percentiles; the dotted orange line indicates the market ATM IV (44.6%). Semi-transparent blue paths show individual scenario trajectories with stochastic fluctuations and occasional tail-state-driven spikes.

Table 5: American option price distributions from 500 forward scenarios (NVDA, evaluated at $t = 5$). Strike levels are expressed relative to the spot price $S = \$207$.

Type	Strike	DTE	Mean ($)	Std ($)	90% CI ($)
Put	186 ($0.90S$)	80	12.59	4.99	[5.6, 21.1]
Put	197 ($0.95S$)	80	16.69	5.56	[8.5, 26.2]
Call	207 (ATM)	80	25.59	6.64	[15.5, 37.0]
Call	217 ($1.05S$)	80	20.94	6.29	[11.4, 31.7]
Call	228 ($1.10S$)	80	16.72	5.81	[8.1, 26.6]

2.31% IV across seven expirations), confirming that the term-structure geometry could be identified from a single cross-sectional snapshot. The equilibrium initialization $v_0 = \theta(t{=}0)$ eliminated v_0 as a free parameter and produced the smile, skew, and term structure by construction.

The multi-expiration calibration resolved the earlier limitation of fixing β_1, but revealed that the ATM term structure was not a simple power law; the U-shape required a quadratic extension (β_5). The fitted $\beta_1 < 0$ contradicted the initial contango assumption ($\beta_1 = 0.20$), demonstrating that the dominant short-term effect was IV *decay* from gamma compression rather than contango. Whether this U-shaped structure persists across market regimes or flattens during low-volatility environments remains an open question that time-series calibration could address. The shared $(\beta_2, \beta_3, \beta_4)$ from NVDA did not perfectly transfer to all tickers (RMSE increased from 1.2% for NVDA to 4.7% for INTC), and the residual analysis (Figure 1c) revealed systematic patterns in the wings, suggesting that per-ticker smile calibration would be necessary for applications requiring sub-2% IV accuracy across all names. More broadly, the current calibration used single-date snapshots; time-series calibration fitting θ_{s_t} values across multiple dates with known HMM states would produce a more robust mapping from regime states to variance levels and would allow the mood sensitivity γ to be identified from data rather than set heuristically.

Several natural extensions follow from these results. The multi-expiration calibration used NVDA data only; collecting cross-sectional snapshots for additional tickers would determine whether the term-structure parameters (β_1, β_5) are universal or require per-ticker fitting, analogous to the smile-shape parameters. The primary intended use of this framework is generating realistic synthetic training data for machine-learning models in finance, with applications ranging from training option

price predictors and learning delta-hedging policies via reinforcement learning to augmenting scarce real option data for transfer learning. The framework naturally supports computation of Greeks (Δ, Γ, $\mathcal{V}$, Θ) via finite differences on the CRR tree, and including these in the synthetic dataset would support hedging applications. Finally, the current mood mechanism created correlated IV dynamics through a shared stress signal; a richer model could introduce ticker-pair IV correlations through the copula structure, producing a full cross-asset IV surface that respects observed correlation patterns between individual-name and index volatility.

7 Conclusion

In this study, we developed a framework for generating synthetic American option prices with self-consistent implied volatility dynamics. The framework linked a Jump Hidden Markov Model to a modified Heston stochastic variance process via a state-dependent mean-reversion target $\theta(t)$, and priced the resulting IV through a CRR binomial tree with early exercise. The equilibrium initialization $v_0 = \theta(t{=}0)$ eliminated the initial IV surface as a free parameter, and the endogenous market mood signal produced correlated IV spikes during stress events without external data.

We calibrated a five-parameter smile and term-structure function ψ on NVDA options, first demonstrating that the smile shape generalized across AMD, MU, and INTC with cross-ticker RMSE of 1–5% IV using only a scalar level adjustment per ticker. Multi-expiration calibration across seven expirations (0–46 DTE) revealed a U-shaped ATM term structure driven by gamma compression at short maturities and uncertainty premia at longer horizons; the quadratic DTE extension (β_5) captured this shape with an overall RMSE of 1.15% IV across the 2–46 DTE range. Forward scenario analysis confirmed that the generated IV dynamics exhibited realistic stochastic fluctuations and a crash-protection premium in OTM puts under stress (1.8–$4.3\times$ calm-to-stress price ratios).

The framework is implemented as an open-source Julia package and is available at `https://github.com/varnerlab/heston-implied-volatility-model`.

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